

# Pharmacology for 2<sup>nd</sup> year HO & Nursing Stud.

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# **Course Outlines**

- ♥ **General Pharmacology**
- ♥ **Pharmacology of ANS**
- ♥ **Cardiovascular (CVS) Pharmacology**
  - ♥ **Blood Pharmacology (Assignment)**
- ♥ **Gastrointestinal & Respiratory Pharmacology**
- ♥ **Pharmacology of Central Nervous System (CNS )**
- ♥ **Endocrine Pharmacology**
- ♥ **Chemotherapy**

# General Pharmacology

# Introduction

## Pharmacology

- It was born in the mid-19<sup>th</sup> century
- Comes from the Greek words “**pharmakon**”, meaning a drug or medicine and “**logos**” meaning discourse
- **Pharmacology** - is the science of the interaction of chemical agents (drugs) with living systems.

# Pharmacology.....

- It encompasses the study of the biochemical and physiologic aspects of drug effects, including absorption, distribution, metabolism, excretion, toxicity, and specific mechanisms of drug action.
- Embraces knowledge of the sources, chemical properties, biological effects & therapeutic uses of **drugs**.
- Pharmacology is the **basis** of much of the research and development of new **drugs**.

- **Drug** - a substance (chemical agent) that affects a biological system in a potentially useful way.
  - Drugs are used in the prevention, diagnosis, treatment or cure of disease or palliation in man or other animals.
- **Poison** - Any chemical agent that produces a harmful effect
- **Toxin** - a poison of biological origin (e.g. tetrodotoxin)

# Principles & Branches of Pharmacology

## Two Basic Principles of Pharmacology

- I. **Pharmacodynamics** - The study of the relationship between concentrations of drug and the biologic effects (physiological or biochemical) with time.
  - In operational terms “**what drugs do to the body**”.
- II. **Pharmacokinetics** - The study of the absorption, distribution, metabolism and excretion of drugs from the body.
  - In operational terms “**what the body does to drugs**”.

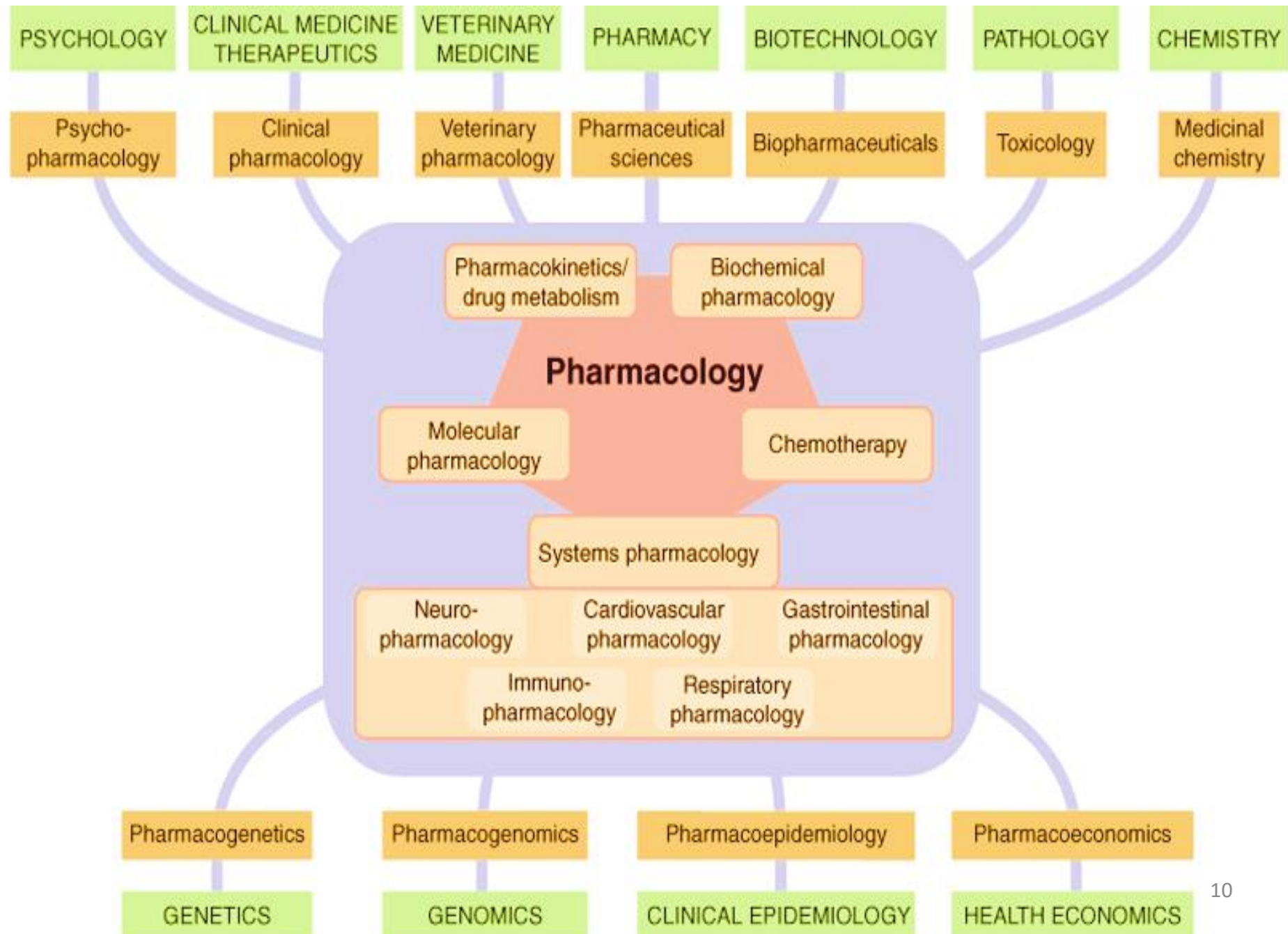
# Branches of pharmacology

## ➤ Toxicology

- is the science of poisons
  - It deals with the adverse effects of drugs and poisonous effects of various chemicals (household, env'tal, industrial or homicidal).
  - It is also concerned with their source, chemical composition, action, tests for detection and **antidotes**.

- **Clinical Pharmacology and Therapeutics**
  - This discipline is interested in characterizing the indications and therapeutic uses of drugs.
- **Pharmacogenomics-** deals with **genetically mediated variations** in drug responses
- **Neuropharmacology** –drugs on NS (neurons)

# Other branches & Relationship with other disciplines



# IMPORTANCE OF PHARMACOLOGY

- Understand drugs and how they can affect living human being
- Know the right dosage form of drugs
- Identify and respond to **drug interactions**, **reactions** and **side effects** and treat accordingly
- Know when to use drugs b/c some conditions do not need drug therapy
- Understand the process of drug intake, absorption, distribution, metabolism and excretion.

# IMPORTANCE OF PHARMACOLOGY...

- Identify the properties of drugs
- Know the application of pharmacology with regards to the five R's of;
  - Use **right** drug
  - Give to **right** patient
  - Give **right** dose
  - Give by **right** route
  - Give at **right** time

# SOURCES OF DRUGS

## NATURAL SOURCES

- **Plant:** from roots, leaves or barks of plants such as **atropine** from *Atropa belladonna*, quinine from *Cinchona* bark, digitoxin, **digoxin**, clove oil, peppermint oil, eucalyptus oil etc.
- **Animal:** Insulin, Thyroid powder, Heparin, Hormones and vitamins
- **Microbiological:** penicillin, chloramphenicol, griseofulvin, neomycin and streptomycin
- **Mineral Sources:** Ferrous sulfate, Magnesium sulfate, Magnesium trisilicate, Aluminium hydroxide

# SOURCES OF DRUGS..

## Synthetic Sources

- Most medications--- ciprofloxacin

## Semisynthetic Sources

- Semi-synthetic human insulin and ampicillin, amoxicillin

## Biosynthetic Sources (Genetically Engineered Drugs)

- Recombinex HB – a hepatitis – B vaccine
- Humulin – human insulin for diabetes

# Drug Nomenclature

## Three types of naming system

### Chemical name

- ✓ It describes the chemical/molecular structure of the drug
- ✓ Chemical names are too complex and cumbersome to be used in prescription
- ✓ E.g. Atenolol: 4-[2-hydroxy-3-[(1-methyl ethyl)amino] propoxy] benzene acetamide.

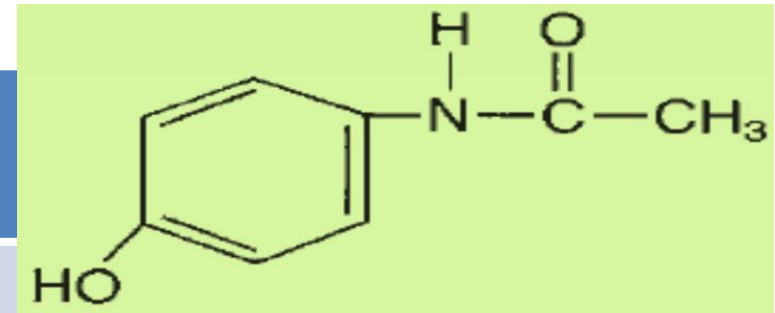
### Generic name

- ✓ Non proprietary name
  - ✓ Internationally accepted name
  - ✓ Assigned by international organization
- E.g. Propranolol, Omeprazole, paracetamol, atenolol etc

## Brand name/trade name

- ✓ Proprietary name assigned by the manufacturer (s)
- ✓ One drug may have several Proprietary name e.g.

| Type of drug name | Examples                                                                                                                                                   |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical name     | N-acetyl-para-aminophenol                                                                                                                                  |
| Generic name      | Paracetamol, acetaminophen                                                                                                                                 |
| Trade name        | Tylenol <sup>®</sup> , Paramol <sup>®</sup> , Rectol <sup>®</sup> , Adol <sup>®</sup> , kelvin <sup>®</sup> , Panadol <sup>®</sup> , Panadrex <sup>®</sup> |



✓ Which naming is preferred in prescribing drugs???

# CATEGORIES OF DRUGS

## ✓ Prescription Only Drugs

- Used under medical supervision
- They are dispensed only by an order of registered physician.
  - E.g. Antibiotics, narcotics, psychotropics

## ✓ Non-Prescription Drugs

- Can be dispensed without physician's prescription
- Over the counter (OTC)
  - E.g. Vitamins, antacids, painkillers etc.

# Dosage forms of drugs

## Dosage form:

- Is the form of preparation designed for administration to the body
- Physical form of a drug

## Types

- Liquid
- Semi solid
- Solid
- Inhalation

## Liquids

- Solutions
- Suspensions
- Emulsions

## **Semi solids**

- Ointments
- Creams
- Pastes
- Jellies

## **Gases**

Inhalation-oral/nasal

## **Solids**

- Powders
- Granules
- Tablets
- Capsules
- Suppositories

# Routes of drug administration

- ☞ Drugs are administered for either their action in the **locality** of their administration or for general **systemic** purpose.
- ☞ Many factors dictate the route
  - The nature of the target tissue
  - Physicochemical properties of the drug
  - The disease state etc..

# Routes drug of administration...

## ➤ Enteral route

- Oral
- Rectal
- Sublingual
- Buccal

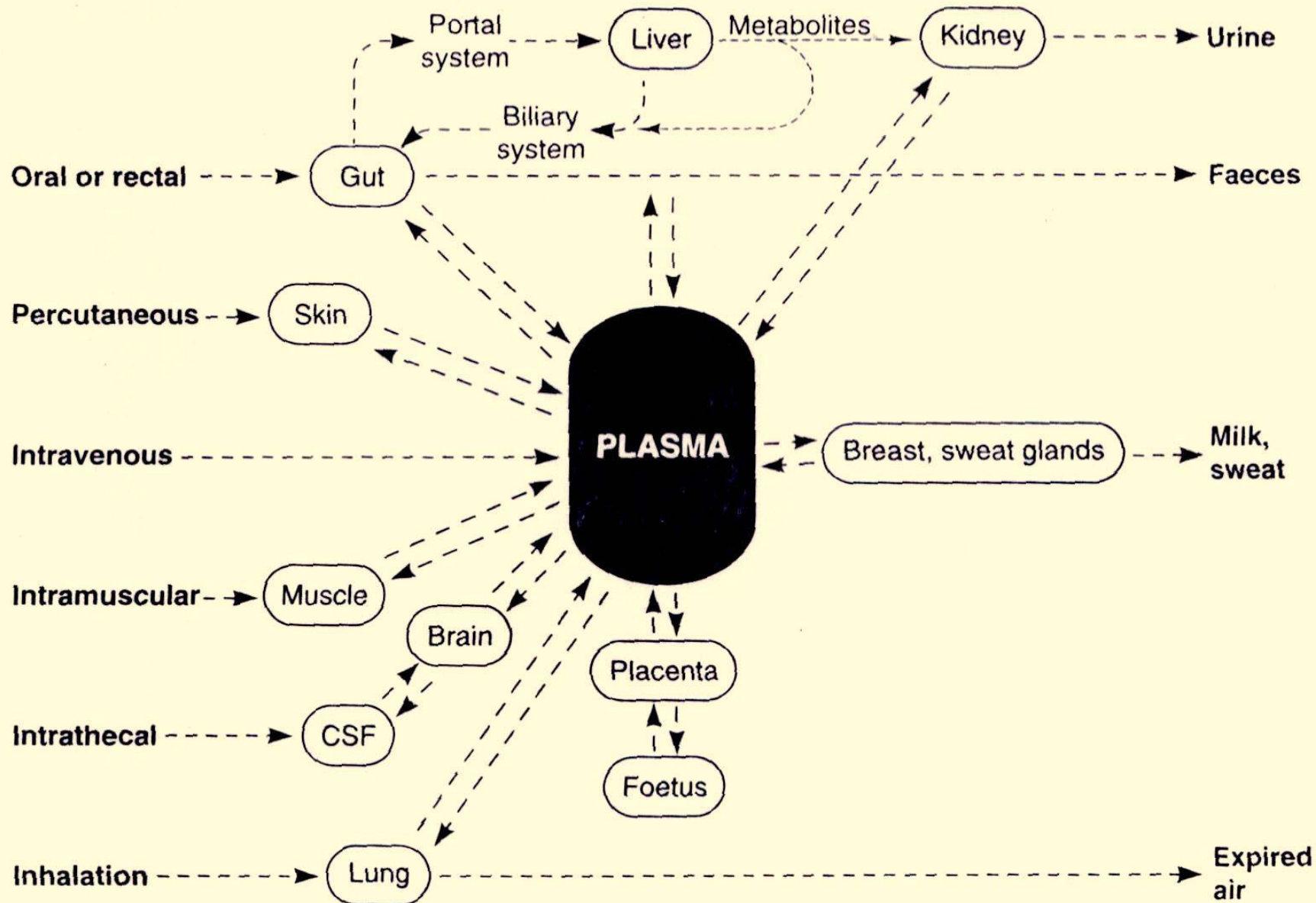
## ➤ Parenteral

- IM, ID, SC, IV

## ➤ Topical

- Skin
- Eye
- Mucous membrane (nasal, rectal and vaginal)

## ➤ Inhalation

**Administration****Absorption and distribution****Elimination**

# Oral ingestion

- ☞ Refers to the administration of drugs through the mouth, **mostly**, for systemic effect.
- ☞ It is the commonest route of drug administration
- ☞ Advantages
  - Safe, more convenient and economical
  - Often painless, need no assistance for administration
  - Both solid and liquid dosage forms can be administered

# Oral ingestion....

## Disadvantages

- Limited absorption of some drugs because of their physical characteristics (e.g., low water solubility or poor membrane permeability)
- Action slower and thus not suitable for emergencies
- Unpalatable drugs difficult to administer
- Not suitable for uncooperative, unconscious & vomiting patients

# Oral ingestion...

## Disadvantages...

- Irregularities in absorption in the presence of food or other drugs
- Destruction of some drugs by digestive enzymes or low gastric pH
- Drugs in the GIT may be metabolized by the enzymes of the intestinal flora, mucosa, or liver **before** they gain access to the general circulation (**1<sup>st</sup> pass metabolism**)

# Sublingual administration

- ☞ Some drugs are taken as smaller tablets which are held **under the tongue**.
- ☞ Nonpolar/ lipid soluble drugs are rapidly absorbed      E.g. Nitroglycerine
- ☞ Drug is protected from **first pass metabolism** as there is direct drainage into the superior vena cava.

## Disadvantages

- ☞ Inconvenient & suitable **only** for small doses
- ☞ unpleasant taste of some drugs

# Parental route

- ☞ **Par = beyond and enteral = intestine**
- ☞ Drug directly introduced into tissue fluids or blood without having to cross the intestinal mucosa
- **This route is used when:**
  - Drug is poorly absorbed from the GIT
  - Drug is destroyed in GIT
  - Drugs which are highly GI irritant
  - Rapid absorption of drug is essential in **emergency**
  - The patient is uncooperative or unconscious

# Parenteral route ...

## Examples of parenteral route of administration

- A. Intra-muscular (IM)-deep into the skeletal muscles (deltoid, gluteous)
- B. Intravenous (IV) - either bolus or infusion
- C. Subcutaneous (SC) - injected into subcutaneous tissue and only non-irritant drugs can be given.  
  
Also important for implantation of solid pellets.
- D. The less commonly used parenteral route of drug administration
  - Intra-peritoneal, Intra-pleural, Intra-arterial, Intra-theccal

# Parental route ...

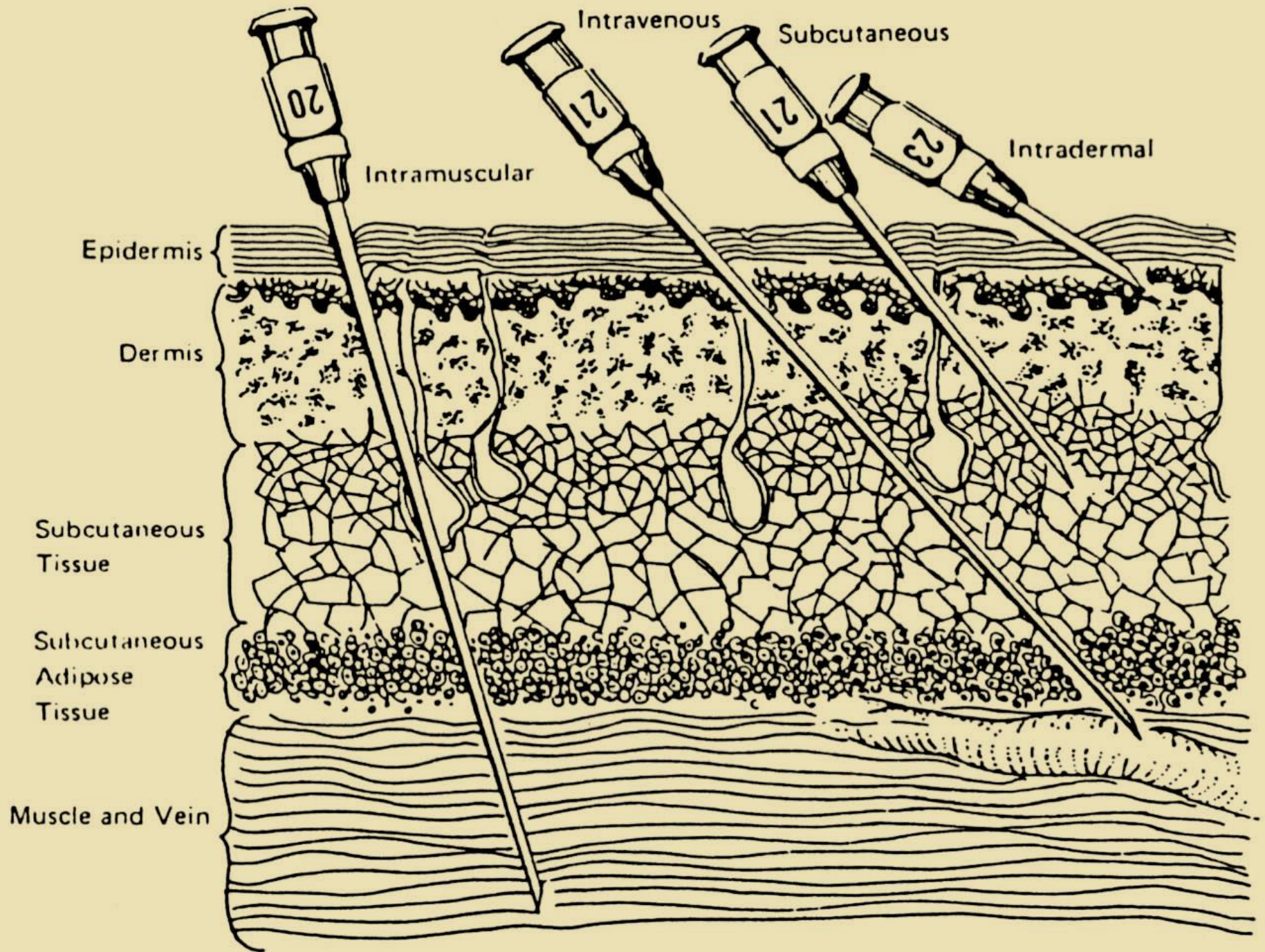
## Advantages

- More rapid response (emergency)
- More accurate dosage
- Useful in vomiting and unconsciousness
- No interference of food or digestive juice
- First pass effect is **bypassed** to a certain extent

# Parenteral route ...

## Disadvantages

- Because of rapid onset, little time to reverse ADR and accidental overdoses
- Sterile dosage forms
- Aseptic procedures
- May be painful
- Relatively expensive
- Patients can not usually administer to themselves

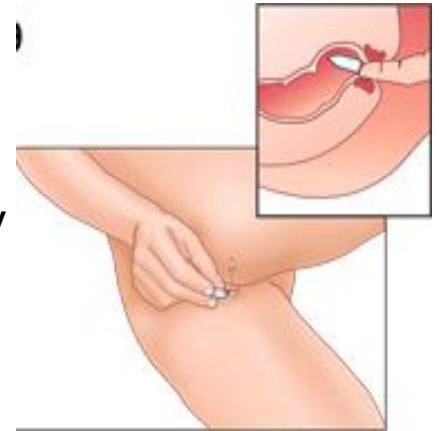


# Rectal administration

- The rectal route often is useful when:
  - ✓ patient is unconscious & vomiting is present

## ➤ Advantage

- For irritating drug e.g. indomethacin suppository
- To produce local effect
- For unconscious patient
- For children e.g. paracetamol suppository
- **Less** first pass metabolism than oral route



# Rectal administration

## ➤ Disadvantage

- Inconvenience of administration
- Slow absorption of drugs
- Irregular and incomplete absorption
- Many drugs can cause irritation of the rectal mucosa

# Vaginal Route

- Drug may be administered locally in the vagina in the form of pessaries. E.g. Antifungal (miconazole) vaginal pessaries

## ➤ Advantages

- Easiness of administration
- Possibility of self-administration
- Bypass/ avoid hepatic first pass-effect
- Low systemic drug exposure for local conditions

# Vaginal Routes ...

## Limitation

-  Social taboos, unawareness and gender-specificity
-  Menstrual cycle-associated vaginal changes
-  Genital hygiene issues
-  Local side effects
-  Coitus interference
-  Variable drug permeability

# Inhalation

- The drug prime particle will leave capillary area of the alveoli of the lungs, which provide an excellent absorbing surface for volatile drugs, gases.
- The drug will pass the mucus membrane of the respiratory system for systemic effect.
- Very efficient way of administering of drug
  - Eg- aerosol, nebulizer; salbuamol, beclomethasone puff

# Inhalation...

- ☞ Suitable for gaseous and volatile drugs
- ☞ Rapid absorption due to large surface area of alveoli
- ☞ Solutions can be atomized and the fine droplets in air (aerosol) inhaled
- ☞ Advantage: Rapid absorption, avoidance of hepatic first pass loss, local application to the pulmonary system.
- ☞ Disadvantage: Poor ability to regulate the dose and irritation of the pulmonary mucosa

# Topical administration

☞ Application could be on mucous membranes, skin or the eye.

## ☞ Mucous membranes

➤ Drugs are applied on the mucous membranes of sublingual, nasal, rectal and vaginal

# Topical administration...

## Skin

- Absorption is proportional to the surface area and lipid solubility of the drug.
- Conditions that increase cutaneous blood flow enhance absorption.
- Systemic absorption from skin is sometimes a reason of toxicity.

# Transdermal

## Transdermal patch

- A medicated adhesive patch applied to skin
- \* Slow effect (prolonged drug action)
- \* Produce systemic effect
- e.g. nicotine patches, scopolamine patches, Nitroglycerin patches



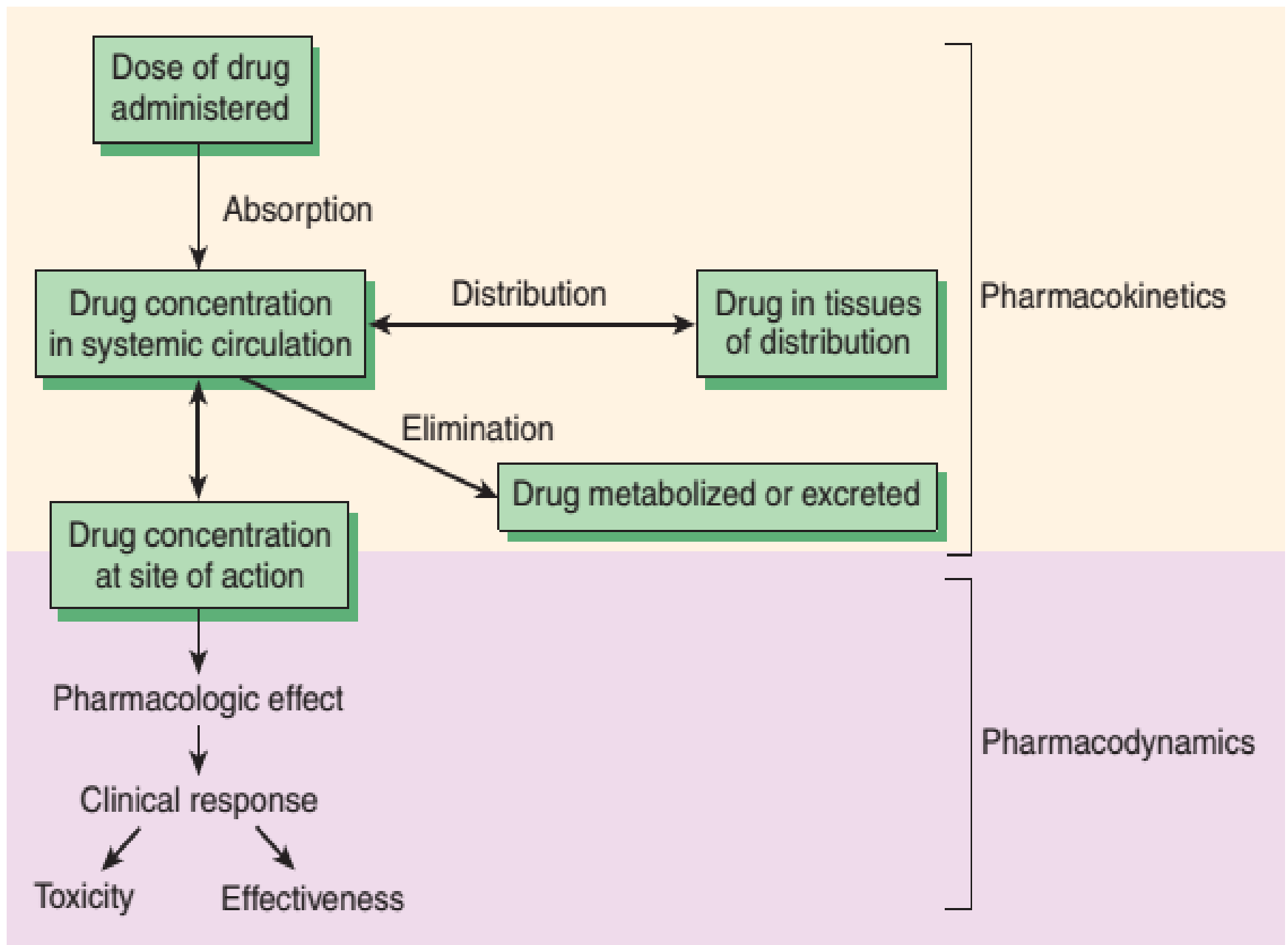
## Eye/opthalmic

- Ophthalmic preparations are meant for their local action
- Systemic absorption that results from drainage through the nasolacrimal canal is usually **undesirable**
- Very little is lost through drainage; hence, systemic side effects are minimized

# *Pharmacokinetics*

# PHARMACOKINETICS

- It deals with the fundamental processes that determine the concentration of a drug at any moment and in any region of the body
- From two words
  - ✓“Pharmacon” ➡ drugs or poisons
  - ✓“Kinesis” ➡ motion
- The 4 major pharmacokinetic processes are:
  - ✓Drug absorption
  - ✓Drug distribution
  - ✓Drug metabolism
  - ✓Drug excretion



# Absorption

- Absorption is the process by which a drug enters the bloodstream without being chemically altered.
- The common sites of drug absorption:
  - **Skin**
  - **Mucous membrane:** sublingual, nasal, rectal and vaginal
  - **Stomach**
  - **Intestine**
  - **Lung cells**
- All the pharmacokinetic processes involve passage of drug through membranes

# Absorption...

## Cell membranes

- Are bilayer of amphipathic lipids, with their hydrocarbon chains oriented inward to form a continuous hydrophobic phase and their hydrophilic heads oriented outwards.
- Membrane proteins embedded in the bilayer serve as **structural anchors, receptors, ion channels, or transporters** to transduce electrical or chemical signaling pathways and provide selective targets for drug actions

# Absorption...

➤ Passage of drugs through cell membranes follows one or a combination of the following major mechanisms.

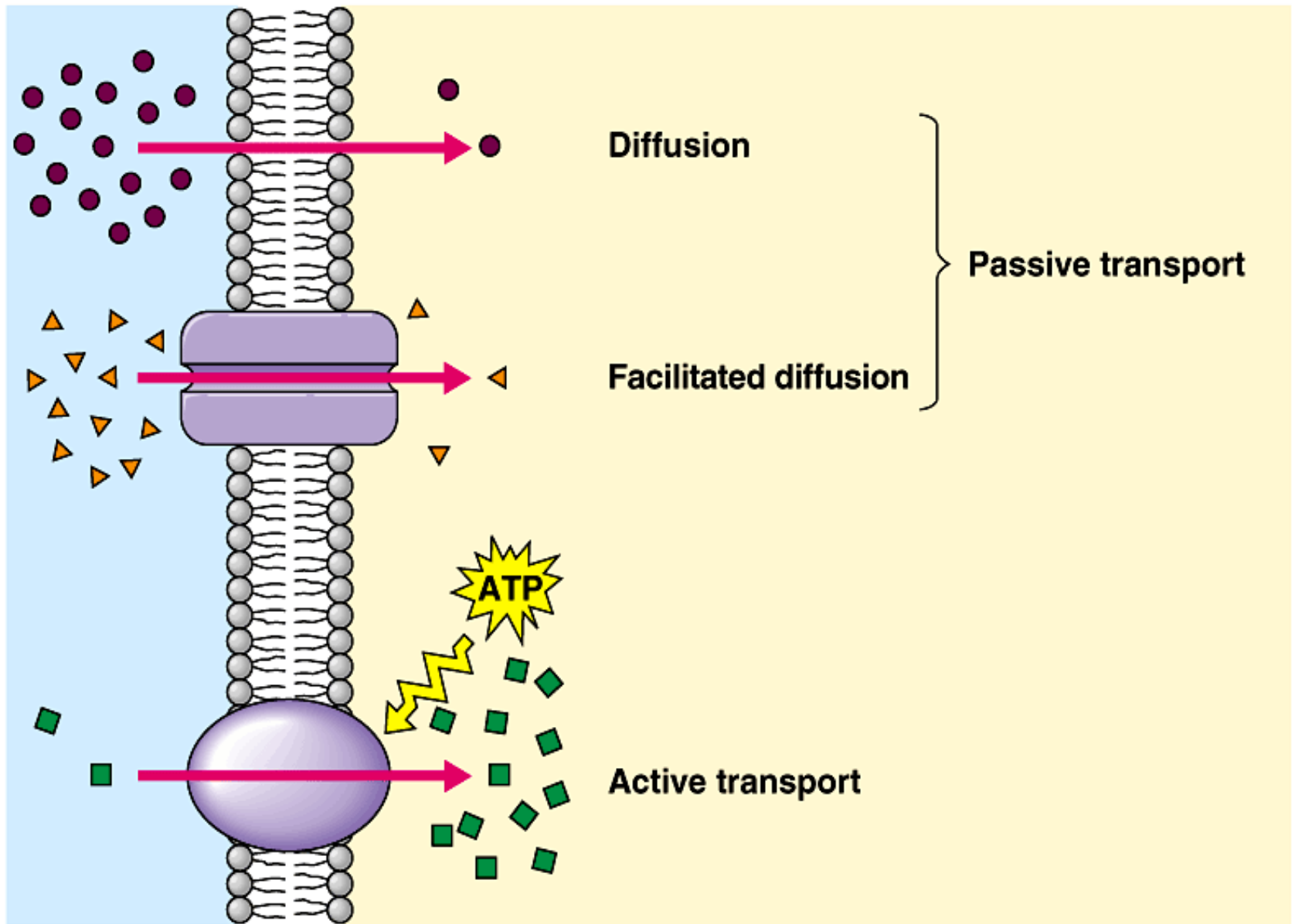
✍ Passive diffusion

✍ Carrier mediated transport

- Facilitated diffusion

- Active transport

✍ Endocytosis



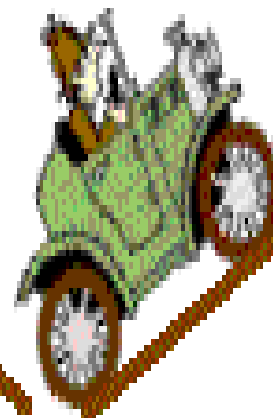
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Passive



Facilitated



Active

# Passive diffusion

- Also called **non-ionic diffusion**, is the **major** (more than 90%) mechanism for absorption of drugs.
- Water-soluble drugs penetrate the cell membrane through aqueous channels or pores, whereas **lipid-soluble** drugs readily move across most biologic membranes due to their solubility in the membrane lipid bilayers.
  - Does not involve carriers
  - not saturable
  - show low structural specificity

# Carrier mediated transport

## a. Facilitated diffusion

- Describes a carrier-mediated transport process in which there is no input of energy.
- Enhanced movement of the involved substance is down a chemical gradient

E.g. levodopa & amino acids into brain

glucose across a muscle cell membrane by GLUT4

# Carrier mediated transport...

## b. Active transport

- ✓ Use ATP & carrier proteins
- ✓ Saturable and structurally specific
- ✓ Against the concentration gradient
- ✓ (e.g. penicillin secretion)
- ✓  $\text{Na}^+/\text{K}^+_{\text{ATPase}}$  is an important example of an active transport mechanism that is a therapeutic target of digoxin in the treatment of heart failure

# Factors that affect drug absorption

- pH (ionization)
- Drug solubility at the site of absorption
- Dosage form
- Contact time at the absorption surface
- Blood flow to the absorption site
- Area of the absorbing surface
- Route of administration

# Factors ...

## ➤PH (Ionization)

- Many drugs are weak acids or weak bases and can exist in either nonionized or ionized forms in an equilibrium, depending on the pH of the environment and the pKa (the pH at which the molecule is 50% ionized and 50% nonionized)
- Only the nonionized (uncharged), lipid soluble, form of a drug crosses biological membranes
  - Weak acids are better absorbed from the stomach
  - Weak bases are better absorbed from the intestine

# Factors ...

## ➤ **Drug solubility at the site of absorption**

- For drugs given in solid form, the rate of dissolution may limit their absorption, especially drugs of low aqueous solubility.
- Ability to diffuse through lipid bilayers (lipid solubility) is important for most drugs
- However, water solubility can influence permeation through aqueous phases.

# Factors ...

- **Blood flow to the absorption site:** The intestines receive much more blood flow than the stomach, so absorption from the intestine is favored over the stomach.

**Note:** Shock severely reduces blood flow to cutaneous tissues, thereby minimizing absorption from SC administration.

# Factors ...

## ➤ **Total surface area available for absorption:**

- Small intestine has **microvillus**; it has absorption surface 1000 times that of stomach.
- Thus, any factor that accelerates gastric emptying will be likely to increase the rate of drug absorption

# Factors ...

## ➤ Contact time at the absorption surface:

- If a drug moves through the GIT very quickly, as can happen with **severe diarrhea**, it is not well absorbed.
- Conversely, anything that delays the transport of the drug from the stomach to the intestine **delays the rate** of absorption of the drug.
  - ✓ The **presence of food** in the stomach both dilutes the drug and slows gastric emptying.
  - ✓ Therefore, a **drug taken with a meal** is generally absorbed more slowly.

# Factors ...

## ➤ Routes of administration

- IV??
- Inhalation- fastest absorption
- IP
- IM
- Sc
- PO
- Topical

Compare their **bioavailability**??

# Bioavailability

- Bioavailability (BA) is the rate and extent to which an administered drug reaches the systemic circulation.
  - For example, if 100 mg of a drug is administered orally and 70 mg is absorbed unchanged, the bioavailability is 0.7 or 70%.
- Determination of bioavailability is important for calculating drug doses for non-intravenous routes of administration.

$$BA = \frac{\text{AUC oral}}{\text{AUC IV}} = \frac{\text{Amount of drug in plasma}}{\text{Actual dose given}}$$

# Bioavailability...

- Intravascular administration (e.g., IV) does not involve absorption, and there is no loss of drug.
  - Bioavailability = 100% = 1
- With extravascular administration (e.g., per oral) , intramuscular [IM] , subcutaneous [SC], inhalation), less than 100% of a dose may reach the systemic circulation because of variations in bioavailability.

## **Factors that influence bioavailability:**

### ➤ **First-pass metabolism:**

- When the drug is rapidly metabolized in the **liver** or gut wall during the initial passage, the amount of **unchanged** drug entering the systemic circulation is decreased.
- This is referred to as **first-pass metabolism**.
- For example, more than 90% of nitroglycerin is cleared during first-pass metabolism.

# Factors....

## ➤ Extent of absorption:

- **Very hydrophilic drugs** are poorly absorbed (unable to cross lipid-rich cell membranes) e.g. Atenolol
- **Extremely lipophilic are also poorly absorbed** (totally insoluble in aqueous body fluids and, therefore, cannot gain access to the surface of cells) e.g. Acyclovir
- For a drug to be readily absorbed, it must be **largely** lipophilic, yet have **some** solubility in aqueous solutions.
- This is one reason why many drugs are either weak acids or weak bases.

# Factors....

## ➤ Chemical instability:

- Penicillin G is unstable in the pH of the gastric contents
- Insulin is destroyed in the GI tract by degradative enzymes.

## ➤ Nature of the drug formulation:

- Particle size, salt form, crystal polymorphism, enteric coatings, and the presence of excipients (such as binders and dispersing agents) can **influence the ease of dissolution** and, therefore, alter the rate of absorption.

# Drug Distribution

- Is the process by which a drug **reversibly** leaves the bloodstream and enters the **interstitial** and **cellular fluids**.
- Distribution also refers to the reversible transfer of drug from one location to another within the body

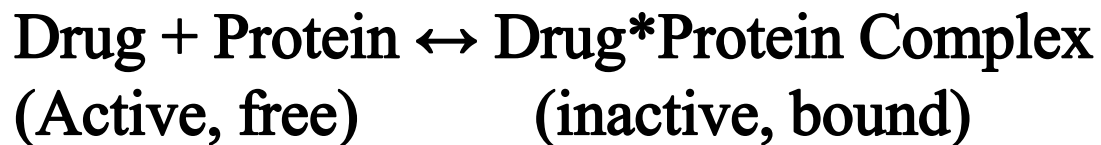
# Factors affecting distribution of drugs

- ❑ Ability to cross cell membrane
- ❑ Plasma protein binding (unbound/ free vs bound drug)
- ❑ Tissue binding
- ❑ Physiological barriers (PB, BBB)
- ❑ Rate of blood flow (Perfusion)

# Factors affecting distribution of drugs

## ➤ Plasma protein binding

- Drugs in the bloodstream move partly as free drug and partly as bound drug, depending on the drug affinity.
- Many drugs circulate in the bloodstream bound to plasma proteins.
- **Only free drug** diffuses into extra vascular space and have pharmacologic as well as toxic effects.
- Bound drugs are pharmacologically inactive



# Factors ....

## ➤ Plasma protein binding ....

- ❖ The binding of drug is reversible and usually non specific
  - Competition between drugs for plasma protein-binding sites may increase the "free fraction" possibly enhancing the effects of the drug displaced.
  - Examples:
    - ⇒ Sulfonamides in a neonate-**kernicterus??**
    - ⇒ Sulphonamide + warfarin ⇒ **bleeding??**
    - ⇒ Salicylates can displace warfarin and oral hypoglycemics

# Factors ....

## ➤ **Tissues Binding**

- ❑ Drug concentrations will vary from organ to organ
- ❑ Can have target tissue therapy or toxicity
  - **Kidney:** Gentamycin
  - **Liver:** Chloroquine
  - **Bone:** TTC, lead & other heavy metals

## Factors ....

### ➤ Physiological barriers

#### ▪ Blood Brain Barrier/BBB

❑ Example: levodopa versus dopamine?

❑ Inflammation increase the permeability of BBB.

E.g:- penicillin and ampicillin – not cross BBB ( highly ionized).

▪ The **placental barrier/PB** prevent transport drugs/chemicals from the mom to the fetus.

**N.B.** For adequate passage of the BBB/PB, dugs should be unionized, have small molecular weight,& lipophilic

# Drug Metabolism (Biotransformation)

- Drug biotransformation
  - Enzyme mediated alteration of drug structure
- The general **objective/ purpose** of biotransformation
  - To facilitate excretion of drugs by rendering them
    - More polar (water soluble) or
    - By conjugating it with highly polar molecules
  - However, Some drugs → Active metabolite(s) & Pro-drug  
→ Active drug

# Outcomes of Biotransformation

- Drug → Inactive metabolite(s) (**Most Drugs**)----- 1<sup>o</sup> goal of metabolism
  - Phenobarbital → Hydroxypentobarbital
- Pro-drug → Active drug
  - L-dopa → Dopamine; Prednisone → Prednisolone
- Drug → Active metabolite(s)
  - Terfenadine → Fexofenadine, Diazepam → Oxazepam
- Drug → Toxic metabolite
  - Acetaminophen → N-acetyl-p-benzoquinoneimine (NAPQI)

# Drug metabolizing systems

- Main site-**liver**, but also intestine, plasma.....others

## A. Microsomal enzymes

- Cytochrome P450 enzyme system (CYP450)- major
  - Glucuronyl transferase
  - Dehydrogenase and Hydroxylase
- Found highly concentrated in **liver** endoplasmic reticulum (microsomes).

## B. Non-microsomal enzymes

- Present in the cytoplasm, mitochondria of d/t organs:  
Esterases, Amidases, Hydrolysis & alcohol metabolism by dehydrogenases

# Phases of Biotransformation reactions

## Phase I

- **Functionalization or non-synthetic reactions**
- Introduce a functional group on the parent drug
  - If not excreted rapidly, the products of phase I bio-transformation undergo phase II conjugation reaction.
  - Include **oxidation** (hydroxylation and dealkylation), **reduction** and **hydrolysis**

### ➤ **Cytochrome P450 isozymes**

- major enzyme systems involved in phase I reactions

## Phase I....

- **Cytochrome P450**, designated as CYP, is a superfamily of heme-containing isozymes
- Four isozymes are responsible for the vast majority of P450-catalyzed reactions.
  - CYP3A4/5, CYP2D6, CYP2C8/9, and CYP1A2.
  - ✓ Considerable amounts of CYP3A4 are found in intestinal mucosa, accounting for **first-pass metabolism** of drugs such as chlorpromazine and clonazepam.

## ➤ Phase II (conjugation or synthetic) reactions

- Conjugation with endogenous compounds via the activity of transferases
    - (glucuronic acid, sulfate, AAs, glutathione or acetate)
  - Form highly polar conjugates which are generally inactive and are rapidly excreted in the urine or feces
  - Require energy (ATP) & saturable process
  - May follow phase I or occur directly
- ❑ **Includes: Glucorinidation, methylation, sulfation, acetylation, and glutathion conjugation**

# Factors affecting drug metabolism

## ➤ Genetic makeup

- eg. Slow and rapid acetylation of **INH, Hydralazine, Dapsone**.

## ➤ Age

- Gray baby syndrome (CAF) in Neonates
- Elderly have hepatic enzymes and other organs deteriorate over time.

## ➤ **Diet:** nutrition plays a critical role in the regulation of drug conjugation.

- Eg. Sleeping time produced by barbiturates increase in prolonged malnutrition.

# Factors affecting .....

## ➤ Sex

- Generally, men metabolize faster than women (ex: alcohol, benzodiazepine)

## ➤ Diseases

ex: hepatitis, cardiac failure → ↓ blood flow to the liver

## ➤ Co-administration of drugs that can be enzyme inducers or inhibitors.

# Enzyme induction

- Exposure to certain drugs, diet and environmental factors may increase synthesis of CYP450
- Increased rate of other drugs (self) biotransformation
- Brings about decreased bioavailability of parent drug and increased concentration of metabolite
  - Can result in either decreased activity or toxicity
- Induction of metabolism of ethinylestradiol/ OCP by rifampicin and phenobarbitone
  - Unwanted pregnancy

# Enzyme inhibition

## Mechanisms

- Competition between two or more drugs for the active site of the same metabolizing enzyme
- Lead to a decrease in the metabolism of one of these agents
  - Depending on the relative concentration of each substrate and their affinities for the enzyme.
  - E.g. Cimetidine vs warfarin

## Elimination half-life ( $t_{1/2}$ ):

- The time required to eliminate 50% of a given amount (or to decrease plasma level to 50% of a former level) ---  $t_{1/2}$
- It has two phases i.e. half-life of distribution and half-life of elimination.
- **Used to determine:** dosage interval, time for a drug to be removed from the body and time to reach steady state.

# Drug excretion

- ☞ Is the passage out of a systemically absorbed drug from the body in the form of **metabolites** or **unchanged** drug.
- ☞ The **kidneys** are the main organ of excretion
- ☞ Other routes of drug elimination:
  - Hepato-biliary excretion
  - Lungs: for volatile substances
  - Sweat
  - Saliva
  - Tears

# Time course of drug response

- To achieve the therapeutic objective, the time of drug response must be controlled
- Need to regulate time at which drug responses **start** the time when **most intense** and time when they **ceases**.
  - Onset of action
  - Duration of action

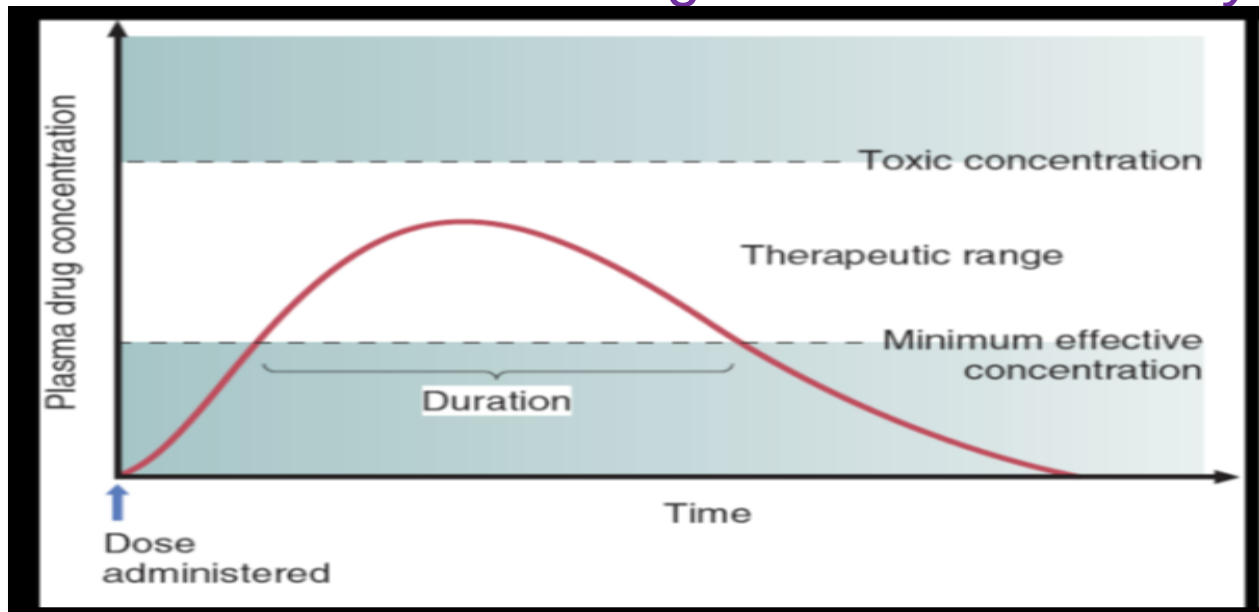
# Plasma drug levels

- **Clinical Significance:**
  - Usually direct correlation between **therapeutic** and **toxic** responses and **plasma drug levels**.
- **Plasma drug levels:**
  - **Minimum effective concentration (MEC):** the minimum plasma drug level below which therapeutic effect will not occur.
  - **Toxic concentration:** the plasma drug level at which toxic effects begins

# Plasma drug levels ...

- **Therapeutic range (window):** the range of plasma drug level falling between MEC and minimal toxic levels (MTD).
  - The objective of drug dosing is to maintain plasma drug levels within therapeutic range (window)

⇒ The case with which drug can be used safely



- Repeated doses produce plateau drug levels (steady state concentration)
  - Elimination equals administered dose
- Techniques to reduce fluctuation in drug level
  - Continuous infusion
  - Reduce dosage size while reducing dosing interval

- **Loading Dose:**
  - A large initial dose when plateau must be reached quickly
- **Maintenance Dose**
  - A dose given to maintain the plateau
  - is a dose administered to maintain the target concentration of a drug.
  - The amount is equivalent to daily excreted dose

# Pharmacodynamics

# Pharmacodynamics

☞ Pharmacodynamics studies the physiological and biochemical effects of drugs and their mechanism of action.

☞ *The study of what drugs do to the body and how they do it.*

☞ Drugs interact with receptors to produce their characteristic effects.

# **Mechanism of drug action (MOA)**

## **Physical means**

- Osmotic action E.g. Osmotic Purgative
- Adsorption E.g. Charcoal in the management of toxin

## **Chemical action**

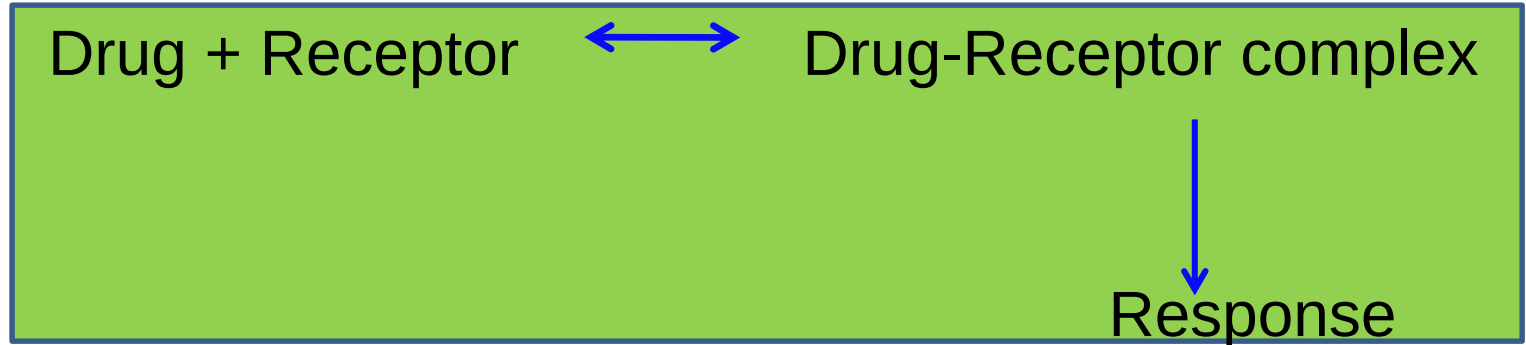
- Antacid to neutralize gastric acid secretions

## **Through protein targets (the commonest)**

- Ion channels, enzymes, receptors, carrier molecules

# Drug-receptor interaction

- Receptors are any functional macromolecules/ proteins in a cell to which a drug binds to produce its effects.



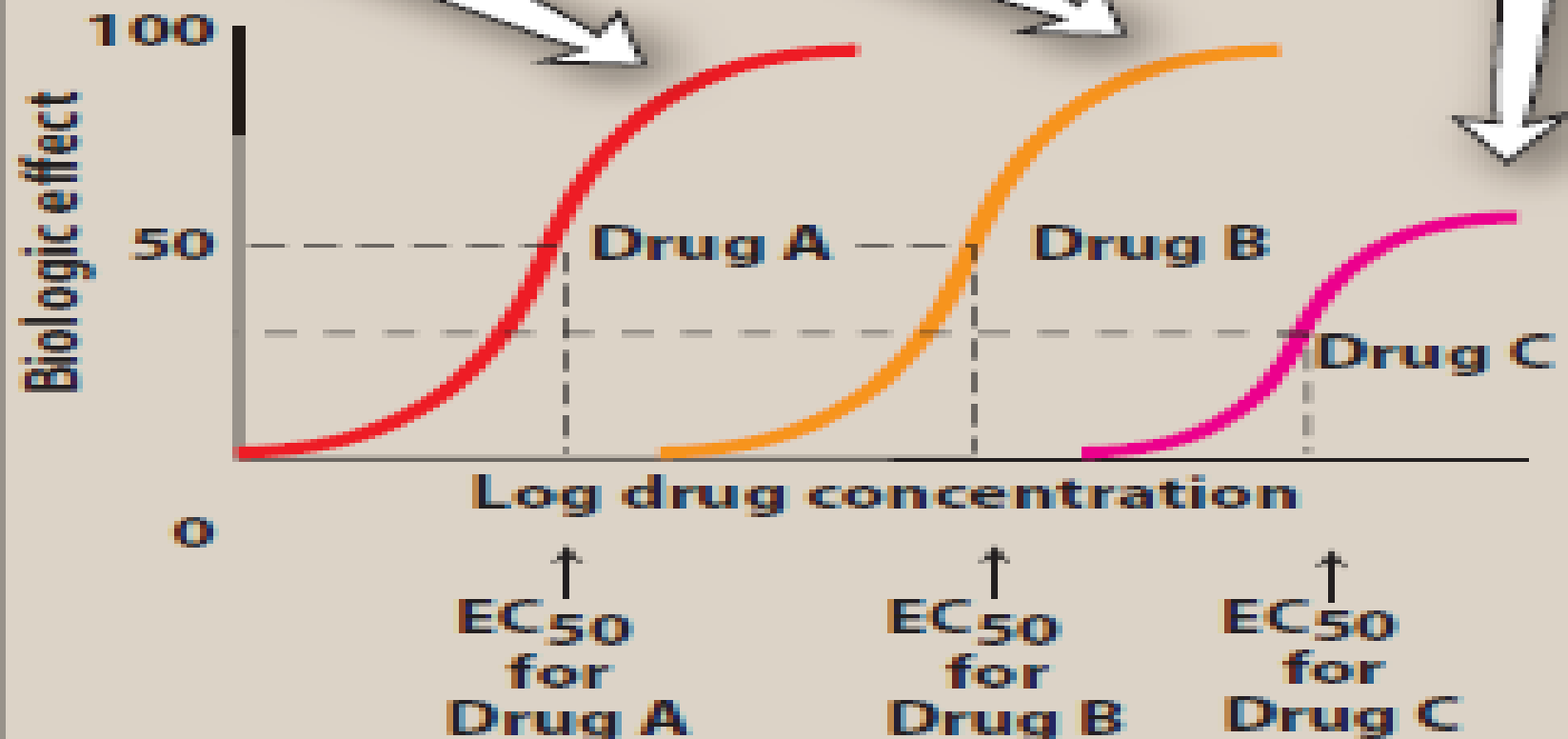
- Under normal condition, receptor activity is regulated by endogenous compounds (NTs, Hormones)
- Drugs mimic or block the actions of EDCs.

# Factors governing drug action

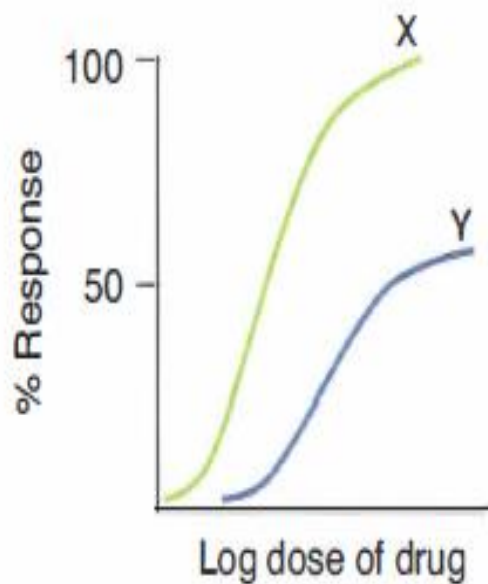
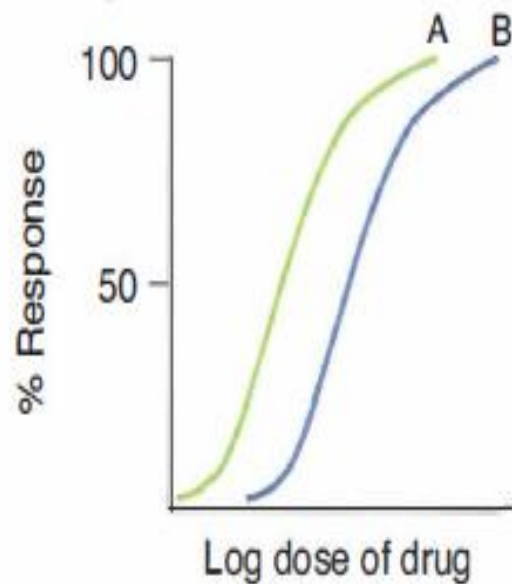
- **Affinity:** ability of drug to bind to receptor
- **Potency:** the amount of drug required to produce an effect of given intensity.
  - Differences in drug potency are evaluated by comparing EC50 (or ED50) values. The lower the EC50????
- **Efficacy:** the ability of a drug to produce a maximum response.
  - Differences in drug efficacy are evaluated by comparing differences in maximal response at high drug doses or concentrations.

Drug A is more potent than Drug B, but both show the same efficacy.

Drug C shows lower potency and lower efficacy than Drugs A and B.



## Parallel and Nonparallel D-R Curves



### Potency

$A > B$

$X > Y$

### Efficacy

A equivalent to B

$X > Y$

- When two drugs interact with the same receptor (same pharmacologic mechanism), the D-R curves will have parallel slopes
- Affinity can be compared only when two drugs bind to the same receptor.
- Drug A has a greater affinity than drug B.

## A. Full agonists

- If a drug binds to a receptor and produces a maximal biologic response that mimics the response to the endogenous ligand
- Full agonists bind to a receptor, stabilizing the receptor in its active state and are said to have an intrinsic activity of 100%
- They have maximal efficacy

## B. Partial agonists

- Partial agonists have intrinsic activities greater than zero but less than one .
- Even if all the receptors are occupied, partial agonists cannot produce the same maximal response as a full agonist.
- When a receptor is exposed to both a partial agonist and a full agonist, the partial agonist may act as an **antagonist** of the full agonist

## D. Antagonists

- Antagonists bind to a receptor with high affinity but possess zero intrinsic activity.
- An antagonist has no effect in the absence of an agonist but can decrease the effect of an agonist when present.
- Antagonism may occur either by blocking the drug's ability to bind to the receptor or by blocking its ability to activate the receptor.

# Antagonism

## I. Competitive antagonists

- Compete with agonists in a **reversible** fashion
- Cause a parallel shift to the right in the D-R curve for agonists
- Can be **reversed** by increasing the dose of the agonist drug
- Appears to decrease the potency of the agonist

# Antagonism

## II. Non-competitive antagonists

### A. Irreversible antagonists

- Irreversible antagonists bind **covalently** to the active site of the receptor
- No shift of EC50
- Reduce agonist efficacy (Decrease Emax)
- **Cannot** be overcome by adding more agonist

### B. Allosteric antagonists:

- Binds to a site (“allosteric site”) other than the agonist-binding site
- No shift of EC50 & Reduce agonist efficacy (decrease Emax).

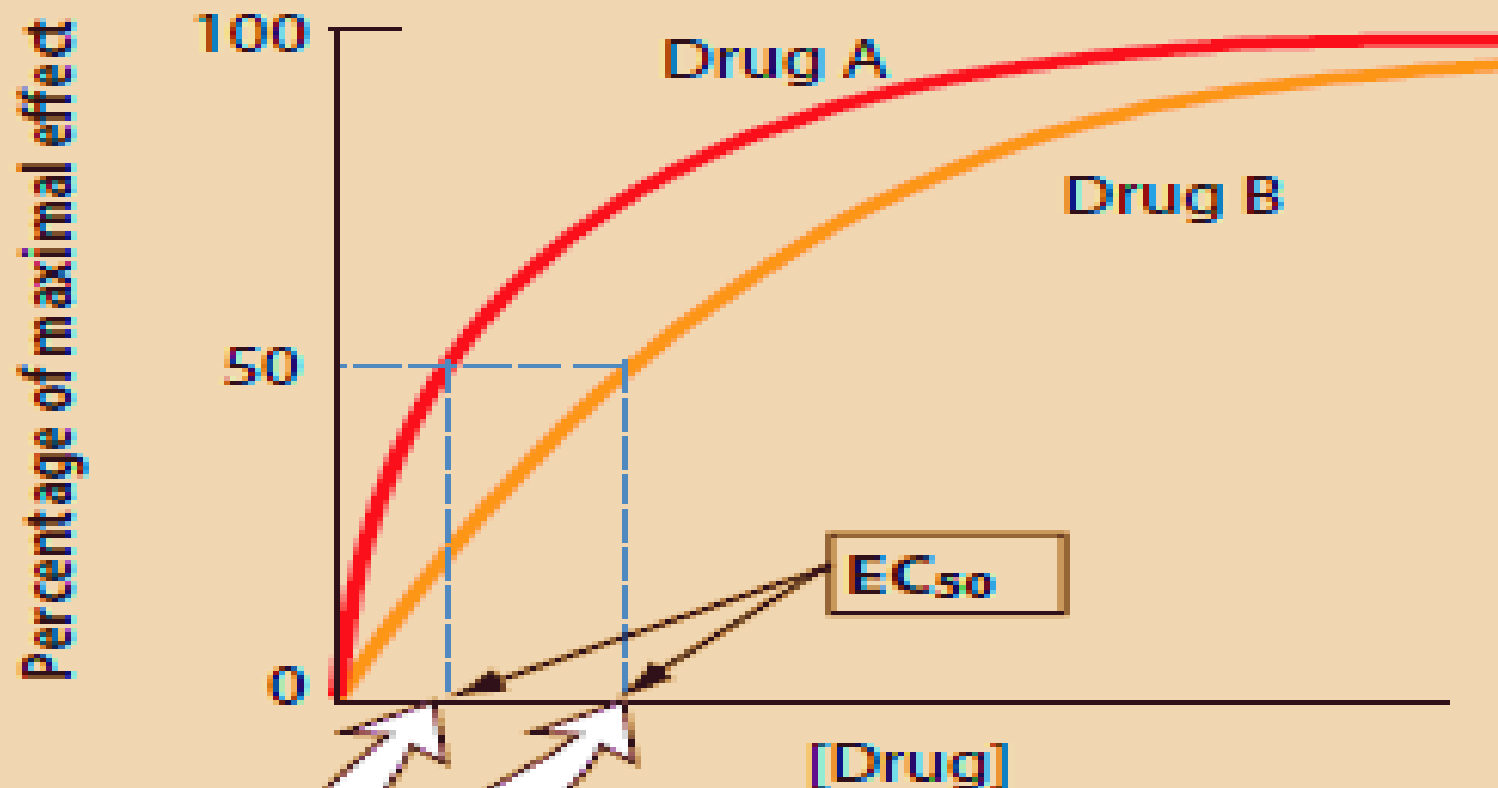
# Dose response relationships

- Graded dose-response
- Quantal response

# Graded dose-response

- As the concentration of a drug increases, its pharmacologic effect also gradually increases until all the receptors are occupied (the maximum effect).
- Plotting the magnitude of response against increasing doses of a drug produces a graded dose-response curve.
- The curve can be described as a rectangular hyperbola
- Two important properties of drugs, **potency** and **efficacy**, can be determined by graded dose-response curves.

# Graded dose response curve



The EC<sub>50</sub> is the concentration of the drug that produces a response equal to 50% of the maximal response.

# Quantal response

- Relationship between the dose of the drug and the proportion of a population that responds to it
- Effect is either present or absent (**all or none**)
- Used for **population** studies
- Commonly employed to study toxicity of a drug

E.g. protection of convulsion by anti-epileptic agents

# Quantal DRCs

- These curves plot the percentage of a population responding to a specified drug effect versus dose or log dose.
- Quantal dose-response curves are useful for **determining doses to which most of the population responds**
- Permit estimations of the median effective dose, or effective dose in 50% of a population-ED50.

- Quantal curves can reveal the range of inter-subject variability in drug response.
  - Steep D-R curves reflect little variability
  - flat D-R curves indicate great variability in patient sensitivity to the effects of a drug.
- **ED<sub>50</sub>** : Dose required at which 50% of those treated show specific intensity of effect (half-maximal effect).
- **LD<sub>50</sub>** : Dose required at which 50% of those treated die (in animals)

# Therapeutic Index

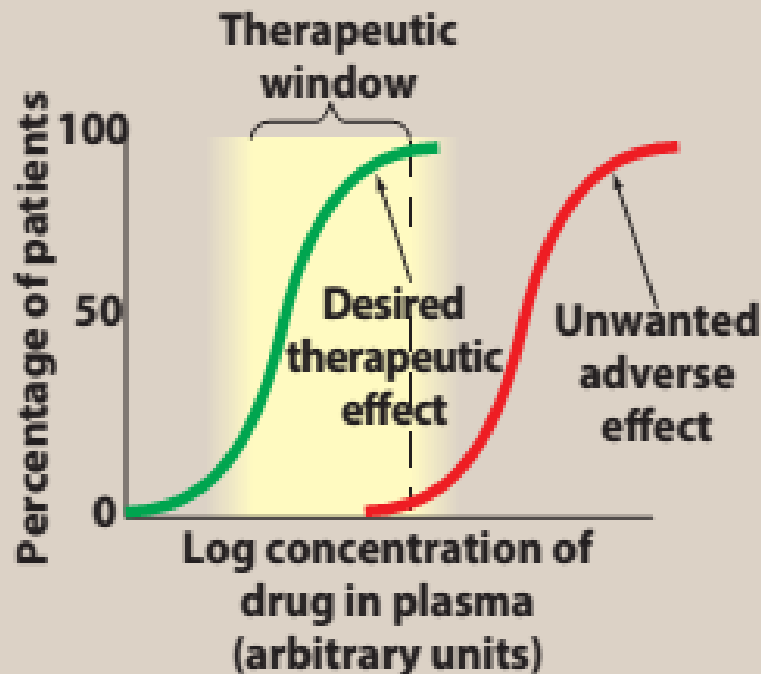
- The therapeutic index (TI) of a drug is the ratio of the dose that produces toxicity in half the population (TD50) to the dose that produces a clinically desired or effective response (ED50) in half the population:

$$TI = TD_{50} / ED_{50}$$

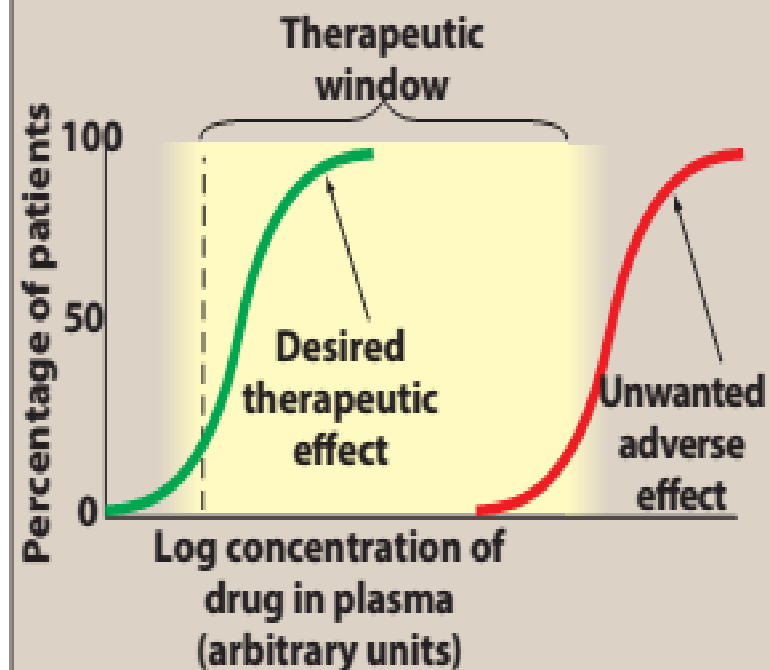
⇒ The TI is a measure of a drug's safety, because a larger value indicates a wide margin between doses that are effective and doses that are toxic.

⇒ In general, a larger T.I. indicates clinically safer drug!

**A** *Warfarin: Small therapeutic index*



**B** *Penicillin: Large therapeutic index*



**Fig: Cumulative percentage of patients responding to plasma levels of warfarin and penicillin**

# Drug interaction

- The modulation of the pharmacological activity of one drug by the prior or concomitant administration of another drug.

⇒ *Enhanced or diminished*

- The knowledge of drug interactions is essential for safe drug therapy.
- Drug interaction could be in in-vitro or in-vivo

## **A. In-vitro interaction**

- ☞ When drugs are added to IV drip or mixed in syringe  
E.g. pencillins & aminoglycosides

# Drug interaction....

## B. In vivo interactions

- Interaction occurs at two stages
  - At pharmacokinetic level
    - At ADME
  - At pharmacodynamic level
    - At the level of receptors and beyond

## **Type of interaction: based on type of agents that interact**

### ❖ Drug –drug interaction

E.g -antacid Vs ketoconazole --- ketoconazole needs an acidic environment for better absorption (↓absorption)

### ❖ Drug-food interaction (TTC Vs milk)

### ❖ Drug –alcohol interaction- Alcohol Vs TCAs

### ❖ Drug-herbal interaction

Based on level of interaction: drug- interaction classified as:-

## **A. Pharmacokinetic level interaction**

### **❑ Absorption**

Cholestyramine +  $\text{FeSO}_4$ : Cholestyramine  $\downarrow$  absorption  $\text{FeSO}_4$

Tetracycline + antacid : interaction in 2 ways

TTC needs an acidic env't for better absorption

Antacid contains divalent metals like  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  since TTC is chelating agent

### **❑ Distribution level interaction**

- High plasma protein binding capacity
- Low excretion rate

Warfarin + phenylbutazone (Warfarin displaced from albumin by phenylbutazone)----  $\uparrow$ se risk of bleeding

### **❑ Biotransformation level interaction**

- ✓ A drug administered may induce or inhibit the liver microsomal enzyme as a result there may be drug interaction in concomitant use of more than 2 drugs e.g. Cimetidine + Warfarin ----- bleeding,

Phenobarbitone + Warfarin  $\rightarrow$  Decreased effect of warfarin

## ❑ **Excretion level interaction**

- Mostly occur at two level:
  - Tubular secretion :- since there is carrier case
    - Eg – probencide + penicillin : since penicillin have short duration of action ----  $\uparrow$ duration ---by  $\downarrow$ excretion
  - Tubular reabsorption: since there is pH based ionization of drug and further degree of reabsorption
    - Eg – weak acidic drug + bicarbonate -----  $\uparrow$  excretion

## **B. Pharmacodynamics interaction**

### **I. Agonistic interaction**

- a. Additive/ summative :  $1+1=2$  ;  
e.g. carbachol + metanichol
- b. Potentiation :  $1+ 0 >1$   
e.g. Amoxicillin+ clavulanic acid (augmentin)
- c. Synergistic :  $2 > 1+1$   
e.g. ampicillin+ gentamicin

# Agonistic Interactions.....

**A. Additive:** the combined effect of two drugs is equal to the sum of the effect of each drug.

**B. Synergistic:** Synergism is said to occur when the combined effects of two agonists exceed that predicted by the individual actions of these compounds (i.e., the resulting effect is more than additive).

Example: Gentamicin + Ampicillin

**C. Potentiation:** The effect of pharmacologically active agent with an agent which doesn't have better effect if given alone.

Example: Amoxicillin + Clavulanic acid

**D. Antagonism:** it occurs when the effect of one drug is diminished by another drug.

A photograph of three bright green apples arranged on a white, circular plate. The apples are positioned in a triangular formation, with one in the foreground and two slightly behind it. The plate is set against a white background. A large, blue, cursive text "Thank You!" is superimposed over the center of the apples. The entire image is framed by a thick black border with a soft orange glow.

*Thank You!*